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The human disease gene LYSET is essential for lysosomal enzyme transport and viral infection

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Lysosomes are key degradative compartments of the cell. Transport to lysosomes relies on GlcNAc-1-phosphotransferase-mediated tagging of soluble enzymes with mannose 6-phosphate (M6P). GlcNAc-1-phosphotransferase deficiency leads to the severe lysosomal storage disorder mucopolidosis II (MLII). Several viruses require lysosomal cathepsins to cleave structural proteins and thus depend on functional GlcNAc-1-phosphotransferase. Here, we used genome-scale CRISPR screens to identify Lysosomal Enzyme Trafficking factor (LYSET) as essential for infection by cathepsin-dependent viruses including SARS-CoV-2. LYSET deficiency resulted in global loss of M6P tagging and mislocalization of GlcNAc-1-phosphotransferase from the Golgi complex to lysosomes. *Lyset* knockout mice exhibited MLII-like phenotypes and human pathogenic LYSET alleles failed to restore lysosomal sorting defects. Thus, LYSET is required for correct functioning of the M6P trafficking machinery, and mutations in LYSET can explain the phenotype of the associated disorder.

Viruses have evolved to hijack the cellular endocytic machinery to enter the cell (1). Lysosomal cathepsin proteases mediate the stepwise proteolytic disassembly of reovirus particles, which is essential for infection (2). Cathepsins also cleave viral glycoproteins of several enveloped viruses during viral entry, triggering productive infection. This includes highly pathogenic viruses including the filovirus family member Ebola virus (EBOV) and the SARS-CoV-1, MERS and SARS-CoV-2 coronaviruses (3–6). These viruses are thus sensitive probes of lysosomal function.

2LYSET is a cellular factor essential for infection by diverse viruses including SARS-CoV-2

To identify genes critical for reovirus infection, we performed a genome-scale CRISPR–Cas9 screen in human glioblastoma cells (U87MG). In these cells, productive viral infection leads to cell death. After CRISPR–Cas9 mutagenesis with the Brunello library (7), which covers 19,114 genes, cells were infected with reovirus type 3D (ReoT3D). To identify protective gene mutations, we

retrieved the guide RNA (gRNA) sequences present in the resistant population and compared them with the unselected population (fig. S1A). In line with its essential role in reovirus entry, the gene encoding cathepsin L (CTSL) was highly enriched in the resistant population (Fig. 1A and table S1) (8). Consistent with the importance of M6P-dependent lysosomal transport for cathepsin activity (9), genes encoding the α/β (GNPTAB (10)) and γ (GNPTG) (11) subunits of GlcNAc-1-phosphotransferase scored highly, as did site-1 protease (S1P), a Golgi-localized protein required for the activation of GlcNAc-1-phosphotransferase (12). We did not identify M6P receptors (MPR), likely owing to functional overlap between the two M6P receptor types, MPRD and MPRI (13). The second highest hit was TMEM251, a gene encoding a largely uncharacterized transmembrane protein. Based on the results outlined below, we renamed this gene **Lysosomal Enzyme Trafficking factor**, *LYSET*. A separate genome-scale CRISPR screen in eHAP cells also identified deletion of *GNPTAB* and *LYSET* as highly protective against reovirus infection, corroborating the essentiality

of M6P-mediated lysosomal protein sorting and LYSET for reovirus infection (fig. S1B and table S2). To validate and characterize the role of LYSET, GNPTAB, and S1P in viral infection, we generated clonal knock out (KO) cell lines in U87MG and 293FT cells using CRISPR/Cas9 (fig. S2). Knock out of LYSET, GNPTAB, and S1P resulted in strong protection against cell death following reovirus infection in both cell types (Fig. 1B). Intracellular viral RNA levels and virus production were severely reduced suggesting an early block in viral entry or replication (Fig. 1, C and D). Because GNPTAB, GNPTG and S1P have known roles in the sorting of lysosomal cathepsins, we reasoned that LYSET might act similarly. We first tested whether LYSET deficiency would prevent viral entry of other cathepsin-dependent viruses (3). As a faithful model of Ebola virus (EBOV) entry (14) we used a GFP-encoding vesicular stomatitis virus pseudotyped with the EBOV glycoprotein (VSV-EBOV-GP). While parental 293FT cells were susceptible to infection with VSV-EBOV-GP, as evidenced by a time-dependent increase in GFP fluorescence, LYSET KO cells were highly refractory to infection (Fig. 1E, fig. S3A, and movies S1 and S2). Similar results were obtained using clonal HAP1 LYSET KO cells and pooled knock outs in human skin fibroblasts (fig. S3, B to D, and movies S3 to S6). SARS-CoV-2 requires activation of its Spike protein during viral entry, which can be mediated by active endosomal/lysosomal cathepsins or by the transmembrane serine protease TMPRSS2 (5). In cells with very low TMPRSS2 expression, cathepsins become essential for SARS-CoV-2 entry. While parental A549-ACE2 cells were highly susceptible to infection by VSV pseudotyped with the SARS-CoV-2 Spike protein, deletion of LYSET strongly reduced infection levels during a time course of infection (fig. S3, E and F, and movies S7 and S8). To validate these results in the context of the authentic virus, we used an infectious molecular clone of SARS-CoV-2 that expresses the mNeonGreen fluorescent protein (15). Consistent with the results of the pseudotyped viruses, we observed a significant decrease in infection (Fig. 1F and fig. S3G). In cells expressing TMPRSS2, most SARS-CoV-2 variants preferentially utilize this route of entry (16). However, the recently emerging omicron variant is less efficiently cleaved by TMPRSS2 and more dependent on cathepsin-mediated entry than other variants of concern such as the delta variant (16, 17). We thus tested lentiviral particles pseudotyped with SARS-CoV-2 S variants in cells that expressed or did not express TMPRSS2. Consistent with our results with Spike corresponding to the early Wuhan strain, LYSET KO resulted in decreased entry of both delta and omicron variant pseudotyped viruses in cells that did not express TMPRSS2 (Fig. 1G). In cells expressing TMPRSS2, the delta variant was only moderately affected by LYSET KO while the omicron variant was still strongly dependent on LYSET (Fig. 1G). Thus, our genetic screens identified LYSET as a host factor essential for a diverse group of viruses that depend on endosomal protease activation during their entry.

LYSET is critical for M6P-mediated lysosomal enzyme transport

The identification of a previously uncharacterized transmembrane protein potentially involved in cathepsin sorting was unexpected, because the key proteins involved in lysosomal targeting are well-established (13). To investigate if LYSET impacted lysosomal trafficking of cathepsins, we analyzed cathepsin B (CTSB) protein levels in 293FT and U87MG wild-type and KO cells (Fig. 2A and fig. S4A). In wild-type cells, the majority of CTSB was present in its mature form following auto-catalytic cleavage in lysosomes. Only low levels were found in the extracellular medium, indicating efficient lysosomal sorting and trafficking. In contrast, LYSET deficiency resulted in aberrant secretion of CTSB precursor forms into the medium concomitant with a near complete loss of mature CTSB in the cell lysate (Fig. 2A and fig. S4A). The dysregulation of cathepsin transport was as pronounced as observed after disruption of core M6P formation components (GNPTAB and S1P). We extended the analysis to a larger panel of cathepsins in 293FT and HAP1 cells (fig. S4, B and C) and measured cathepsin activity using a quenched cell-permeable probe (BMV-109) that covalently links to active cysteine cathepsins and gains fluorescence (18). Compared to wild-type cells, LYSET KO cells displayed a strongly decreased fluorescence after live cell labeling suggesting a global loss of cysteine cathepsin protease activity (Fig. 2B and fig. S4, D and E). This indicated a severe defect in lysosomal protein targeting because activity requires auto-catalytic cleavage in lysosomes. To investigate whether the missorting was specific to cathepsins or pointed to a more general defect, we analyzed the consequences of LYSET deficiency for additional lysosomal proteins and in different cellular contexts. We consistently found large increases in secretion of typical soluble lysosomal proteins into the extracellular medium, while their intracellular protein levels and proteolytic maturation were strongly decreased in KO cells generated in primary human fibroblasts, U87MG, 293FT, HAP1 and SK-MEL-30 cells (fig. S5). In addition, we observed an increase in LC3B type II levels associated with autophagic membranes in cell lysates (fig. S5C). Autophagy markers or the accumulation of an autophagosome-specific dye were markedly elevated in LYSET and GNPTAB KO cells compared to wild-type cells under basal conditions and not further elevated by blocking lysosomal hydrolase activities with chloroquine (fig. S6A) or a combination of chloroquine and rapamycin (fig. S6B). This was expected because the content of autophagic vesicles is degraded by lysosomal hydrolases upon fusion with lysosomes (19). To more comprehensively assess the effect of LYSET knock-out on a primary cell type, we used quantitative proteomic approaches in mouse embryonic fibroblasts (MEFs). We first generated C57BL/6J mice with deletions in *Lyset* by gene editing. We designed sgRNAs to excise a region in the second exon of *Lyset* resulting in two mouse lines with distinct out of frame deletion variants (of $\Delta 184$ and $\Delta 199$ nucleotides) (fig. S7, A to C and

E). To generate mice with *Lyset* gene deletions (referred to as *Lyset* KO), mice were bred as compound heterozygotes ($\Delta 184/\Delta 199$) or homozygotes. We verified loss of *Lyset* protein expression in different tissues (fig. S7D). To investigate the extra-cellular proteome, we collected and concentrated serum-free conditioned medium from wild-type and *Lyset* KO MEFs. In parallel, we prepared cell lysates to determine the intracellular proteome. Using data independent acquisition (DIA) mass spectrometry (fig. S8A) (20), we detected and quantified more than 4,000 proteins in the intracellular proteome and more than 1,000 proteins in the secretome, showing a high correlation between replicates (fig. S8, B to E, and table S3). Only a small subset of proteins were differentially expressed in *Lyset* KO versus wild-type cells. The majority of proteins found in higher abundance in the secretome of *Lyset* KO MEFs were well-characterized luminal lysosomal proteins (indicated in red), while these proteins were depleted from the intracellular proteome (Fig. 2C and fig. S9A). Protein levels of two lysosomal enzymes known to traffic independently from M6P (Gba and Acp2) were unaffected (Fig. 2C), which we confirmed using specific activity assays (fig. S10). Moreover, membrane lysosomal proteins (indicated in blue) were unaffected. Similar results were obtained using parallel reaction monitoring (PRM), a targeted method that provides more sensitive quantification of a smaller subset of lysosomal proteins (fig. S9B) (20). Thus, *LYSET* deficiency results in a severe defect in lysosomal trafficking that is specific for M6P cargoes.

Key steps in lysosomal enzyme sorting are the modification with M6P by GlcNAc-1-phosphotransferase in the Golgi apparatus and the subsequent binding to M6P-specific receptors that mediate the transport to the lysosome (13). To distinguish between an early defect in tagging or a later defect in M6P recognition, we used an unbiased proteomic approach to detect M6P modifications directly on glycoproteins isolated from wild-type cells and cells deficient in *LYSET*. This method, based on immobilized metal ion affinity chromatography (Fe^{3+} -IMAC), allows for enrichment of glycopeptides containing the negatively charged M6P modification followed by glycopeptide identification using mass spectrometry (fig. S11A) (21)). In wild-type 293FT cells we readily identified glycopeptides containing M6P from canonical lysosomal enzymes (Fig. 2D, fig. S11B, and table S4). As expected, these M6P glycopeptides were absent in cells lacking GNPTAB. The *LYSET* KO cells displayed a similar absence of M6P modified glycopeptides. This was confirmed in an orthologous assay using a single-chain M6P-specific antibody fragment (22). Lysates of 293FT and HAP1 cells devoid of *LYSET* showed a large decrease in M6P immunoreactive glycoproteins, comparable to decreases observed with GNPTAB KO and S1P KO cells (Fig. 2E and fig. S11C). Thus, *LYSET* is essential for an early step in lysosomal enzyme transport—the generation of the M6P modification.

LYSET binds to GNPTAB and regulates its function by mediating proper Golgi localization

LYSET is a small, poorly characterized protein containing two transmembrane domains. It co-localizes with GNPTAB in the Golgi apparatus (Fig. 3, A to C, and fig. S12, A to C). We performed immunoprecipitation (IP) followed by immunoblotting experiments to test if *LYSET* interacts with GNPTAB. After co-expression of epitope-tagged *LYSET* and GNPTAB we detected GNPTAB-MYC in the *LYSET*-FLAG IP and *LYSET*-FLAG in the GNPTAB-MYC IP (Fig. 3D). These reciprocal co-IPs appeared to be specific because the endogenous Golgi protein Giantin was not detected, nor were *LYSET* or GNPTAB found in control IPs with epitope-tagged mNeonGreen. This interaction suggests that *LYSET* plays a role in modulating the function of the GlcNAc-1-phosphotransferase complex. Because pathogenic mutations in GNPTAB have been reported that lead to mislocalization of the GlcNAc-1-phosphotransferase complex (23, 24), we hypothesized that *LYSET* might function by retaining GNPTAB in the Golgi apparatus. We examined this by immunofluorescence analysis for endogenous GNPTAB and co-staining with antibodies for the Golgi apparatus (GM130) and lysosomes (LAMP2). The staining pattern in *LYSET* KO HAP1 cells was clearly distinct from wild-type cells. While in wild-type cells GNPTAB was localized predominantly in the Golgi apparatus, *LYSET* KO resulted in a loss of Golgi co-localization and a gain in localization to lysosomal structures (Fig. 4A). To investigate this further, we characterized the expression of the endogenous GlcNAc-1-phosphotransferase complex in subcellular fractions. We used sequential centrifugation to isolate a 20K fraction enriched in lysosomes and a 100K fraction enriched in ER/Golgi membranes (25). *LYSET* deficiency resulted in a near complete loss of endogenous GlcNAc-1-phosphotransferase α -subunit protein levels in the 100K fraction (Fig. 4B). This loss was post-transcriptional because GNPTAB mRNA levels did not substantially differ between wild-type and *LYSET* KO cells (fig. S13). Inhibition of proteasomal degradation with epoxomicin did not rescue expression of GNPTAB in *LYSET* KO cells (fig. S14, A and B). In contrast, preventing lysosomal degradation by blocking organellar acidification [bafilomycin A1 or by protease inhibition (PI; E64d/leupeptin/pepstatin A)] increased GNPTAB protein levels specifically in the lysosome-enriched 20K fraction (Fig. 4C and fig. S14, C to E). Moreover, upon bafilomycin treatment we observed immunoreactive bands of higher electrophoretic mobility likely corresponding to partial cleavage products (Fig. 4C), which was also observed in magnetite-purified lysosomes (fig. S14F). These results suggest that GNPTAB is degraded in lysosomes. Immunofluorescence microscopy in *LYSET* KO cells revealed that the majority of endogenous GNPTAB remained co-localized with a lysosomal marker upon treatment with both inhibitors (fig. S14, G and H). However, in bafilomycin- but not PI-treated cells some GNPTAB co-localized with the Golgi apparatus (fig. S14, G and H). The latter could be because of a disturbance of the pH within the

Golgi apparatus by bafilomycin that could lead to aberrant post Golgi trafficking (26). These data support a model in which LYSET interacts with GNPTAB and is important for proper localization of GNPTAB in Golgi stacks. In the absence of LYSET, GNPTAB is mis-localized to the lysosomes where it is degraded.

LYSET's role in lysosomal transport provides a disease mechanism for a previously described genetic disorder

It has long been recognized that mutations in genes encoding the core components of lysosomal enzyme trafficking cause specific inherited syndromes including mucopolysaccharidosis II (MLII) in which defective GNPTAB mutations are etiological (27, 28). Recently, biallelic LYSET mutations have been described in individuals from two families with a recessive genetic disorder with characteristics of mucopolysaccharidoses/mucopolipidosis (29). However, the mechanistic basis to link LYSET with disease was tentative. Our results suggest that defects in lysosomal protein sorting caused by LYSET deficiency might underlie this genetic disorder. Elevated serum level of lysosomal enzymes is a defining feature of MLII. The measurement of these enzymes in serum is used diagnostically to distinguish MLII from other metabolic diseases that cause similar clinical features (28). Compared to wild-type mice, serum from *Lyset* KO mice showed markedly higher enzyme activities of all tested lysosomal enzymes including α -mannosidase, β -hexosaminidase, α -L-fucosidase, and β -galactosidase (Fig. 5A). Furthermore, electron microscopy (EM) analysis of *Lyset* KO mouse embryonic fibroblasts (MEFs) showed obvious morphological changes of lysosomes (Fig. 5B). Lysosomes with electron dense material were accumulated in the cytoplasm, another characteristic feature of lysosomal storage disorders (28). Quantitative analysis of EM images revealed that lysosomes were significantly more numerous and larger in *Lyset* KO MEFs while the cell area was not significantly different compared to wild-type cells (Fig. 5C and fig. S15, A and B). In line with this, aberrant lysosomes were also observed in human cells lacking LYSET (fig. S16). In addition, *Lyset* KO MEFs were resistant to infection by VSV-EBOV-GP, reinforcing our results in human cell lines (Fig. 5D and movie S9 and S10). Despite showing characteristic MLII phenotypes including increased lysosomal enzyme serum levels and storage materials in lysosomes, *Lyset* KO mice did not display obvious clinical symptoms observed in human patients with LYSET mutations. Similarly, the *Gnptab* KO mouse is not as deleterious as in MLII disease (30).

To link the human patient mutations in LYSET (29) directly to lysosomal protein sorting we first established that lentiviral complementation of wild-type LYSET (isoform 1 and isoform 2) could restore CTSB missorting and maturation in 293FT LYSET KO cells (fig. S17). Subsequently, we tested complementation by LYSET containing the Y72Ter or R45W pathogenic mutations (29). As controls we used wild-type LYSET as well as three nonsynonymous LYSET variants not associated with disease, which are

frequent in the population. R45W displayed slightly lower protein expression levels. As expected, Y72Ter was not detected because this frameshift mutation leads to a premature stop codon. While all controls corrected CTSB missorting, Y72Ter and R45W failed to do so (Fig. 6A). Similarly, R45W expression did not rescue the lysosomal storage defects observed using electron microscopy in LYSET KO cells (fig. S18). Moreover, the pathogenic allele R45W did not rescue the loss of endogenous GNPTAB in 100K fractions (Fig. 6B). Co-immunoprecipitation experiments showed a severe defect of R45W in binding with GNPTAB (Fig. 6C and fig. S19). Because R45W fails to rescue the lysosomal trafficking defect, these results suggest that the interaction of LYSET with GNPTAB is critical for GNPTAB function and that mutations in LYSET that affect this interaction can contribute to disease development. Thus, LYSET deficiency causes defining features of mucopolipidosis II and patient mutant alleles are compromised in the role of LYSET in M6P-dependent lysosomal protein sorting.

Conclusion

Here, we establish LYSET as a Golgi-resident protein essential for M6P-mediated lysosomal protein trafficking. Our results support a model in which LYSET is essential for the activity of GlcNAc-1-phosphotransferase by binding to it and retaining it in the Golgi apparatus. LYSET is relevant to human disease because patients with biallelic LYSET mutations suffer a genetic inherited disorder (29). Based on our elucidation of the role of LYSET in cell physiology, we propose that this disorder is similar to MLII. Because the clinical symptoms of mucopolipidosis and mucopolysaccharidosis overlap and not all cases can be explained by mutations in known disease genes, *LYSET* sequencing may help to identify disease-causing mutations in remaining patients. Furthermore, as an important component of lysosomal function, LYSET is essential for infection by diverse highly pathogenic viruses that rely on endo-lysosomal activation by cathepsins.

Materials and methods summary

A detailed version of the materials and methods is provided in the supplementary materials. In brief, genome-scale CRISPR/Cas9 knockout libraries were generated in eHAP and U87MG cells. Libraries were infected with reovirus type 3D to select for a cell population containing mutations that confer resistance to viral infection. After genomic DNA isolation, PCR was used to amplify sequences encoding guide RNAs and their abundance was quantified using next generation sequencing. Statistical analysis was performed to identify and rank genes that were enriched in the selected population compared to the uninfected population.

LYSET KO mutations were introduced in different human cell lines using CRISPR/Cas9. Cells were infected with reovirus type 3D and viral infection was assessed using crystal violet assay, RT-qPCR and plaque forming assay. For infections with VSV pseudotyped with Ebolavirus GP or SARS-CoV-2 S, viral entry was assayed

using live cell imaging with an Incucyte system to monitor VSV-encoded GFP expression.

The endogenous expression of proteins involved in M6P lysosomal protein transport was determined in extracellular medium, intracellular lysates and subcellular organelle fractions in wild-type and KO cells using immunoblotting, lysosomal enzyme activity assays and mass-spectrometry. Binding between LYSET and GNPTAB was determined using co-immunoprecipitations from lysates of cells transfected with plasmids encoding epitope-tagged proteins.

All experiments involving mice were approved by Stanford's Institutional Animal Care and Use Committee. C57BL/6J zygotes were pronuclear injected with Cas9 RNPs targeting Lyset (Tmem251) with two synthetic gRNAs to introduce a frameshift mutation.

For electron microscopy, cells were fixed, osmicated, and Epon polymerized. Ultrathin sections (60 nm) were prepared and examined in an EM902 microscope. For post-embedding immunogold labeling, ultrathin sections (60 nm) were prepared from cryoprotected cell pellets (2.3 M sucrose), collected on Carbon-Formvar-coated nickel grids, and incubated with one or two antibodies followed by protein A-coupled to colloidal gold particles. Images were acquired with a JEM- 2100Plus Transmission Electron Microscope.

For immunofluorescence, cells treated or not for 24 hours with lysosomal protease inhibitors (Leupeptin, Pepstatin A and E64d) or bafilomycin A1 were fixed with 4% PFA, blocked and permeabilized in BSA/saponin buffer followed by incubations with primary antibodies, Alexa Fluor-coupled secondary antibodies and Hoechst 33342. Images were acquired using a LSM880 confocal microscope equipped with an Airyscan detector. For in situ labeling of glucocerebrosidase (GBA) GBA inhibitory MDW933 was used prior to fixation. BMV-109 probes were applied for live cell imaging of cysteine cathepsin activities on an Axiovert 200 M confocal microscope. Autophagic vacuole accumulation was visualized using the CYTO-ID Autophagy detection kit. All images were processed using ImageJ (<https://imagej.nih.gov/ij/notes.html>). The Manders colocalization coefficient was determined by means of the JACoP plug-in.

For mass-spectrometry, proteins were extracted from cell lysates and conditioned media by chloroform/methanol, digested by trypsin in the presence of RapiGest, and the concentration of desalted samples was determined using a peptide assay. For each sample, 1 µg of peptide was analyzed by UHPLC-MSMS using self-packed columns in combination with a Dionex Ultimate 3000 nano-UHPLC system coupled to an Orbitrap Fusion Lumos mass spectrometer. Data acquisition was performed using parallel reaction monitoring (PRM), with a previously established assay for the analysis of lysosomal proteins, and data independent acquisition (DIA) for both sample types. Raw files were analyzed by Skyline (PRM) and Spectronaut (DIA) with determination of

significantly regulated proteins based on fold-change and p/q-values. The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifier PXD029609.

For glycoproteomics, cells were lysed and protein digestion was performed using an S-trap mini column protocol. Peptides were then enriched with immobilized metal affinity chromatography with iron-functionalized magnetic beads. Subsequently, peptide mixtures were separated over an EasySpray reversed phase LC column. The mobile phases were driven and controlled by a Dionex Ultimate 3000 RPLC nano system. An integrated loading pump was used to load peptides onto a trap column, which was put in line with the analytical column. Eluted peptides were analyzed on an Orbitrap Fusion Tribrid MS system. All data were searched with Byonic using the entire human proteome downloaded and an N-glycan database of 183 N-glycans that included M6P glycans. The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository with the dataset identifier PXD029746.

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Competing interests: J.E.C. consulted for Janssen BioPharma on topics unrelated to this study. C.R.B. is a co-founder and Scientific Advisory Board member of Lycia Therapeutics, Palleon Pharmaceuticals, Enable Bioscience, Redwood Biosciences (a subsidiary of Catalent) and InterVenn Biosciences. C.A.B. reports compensation for consulting and/or SAB membership from Catamaran Bio, DeepCell Inc., Immunebridge, Sangamo Therapeutics, Bicycle Tx, and Revelation Biosciences on topics unrelated to this study. The remaining authors declare no competing interests. **Data and materials availability:** All data presented are available in the main text or the supplementary materials. FASTQ files for CRISPR screens are available at ArrayExpress (Accession# E-MTAB-11158). Acquired glycoproteomics data are available at ProteomeXchange with identifier PXD029746 and DIA and PRM data with identifier PXD029609. Cell lines and LYSET knockout mice generated in the laboratory of J.E.C. are available upon request to J.E.C. under a material transfer agreement from Stanford University. Huh7.5.1 derivatives require an additional material transfer agreement from Apath. **License information:** Copyright © 2022 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works.

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SUPPLEMENTARY MATERIALS

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Materials and Methods

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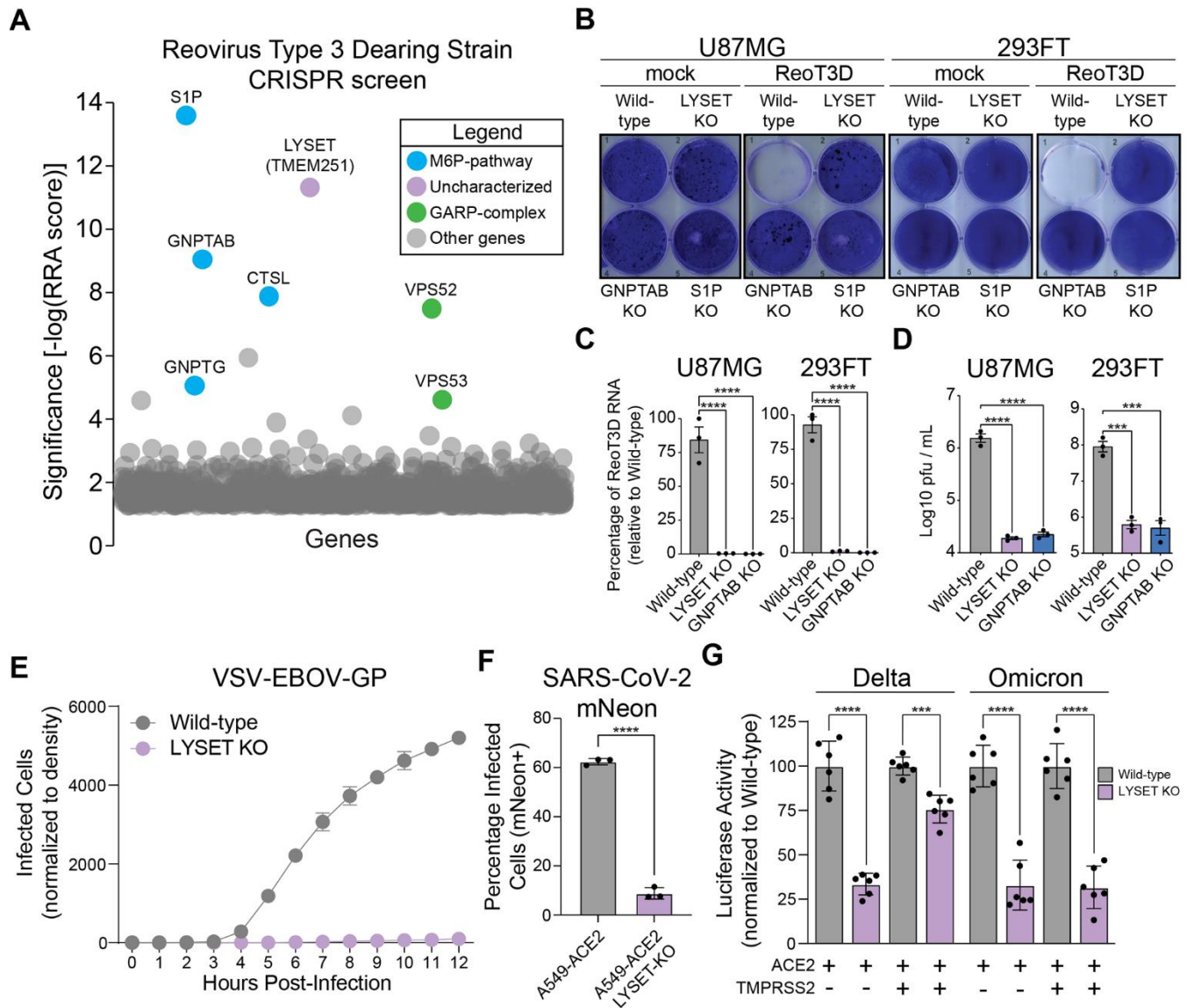


Fig. 1. LYSET is a critical host factor for multiple viruses that utilize activated cathepsins for entry. (A) CRISPR screen for reovirus T3D (ReoT3D) host factors in U87MG cells. Significance of enrichment was determined via MAGeCK analysis (y axis). All genes are plotted as bubbles on the x axis, each representing a specific gene. A subset is color-coded according to function, labels show gene names. (B) Crystal violet staining after infection with ReoT3D representative of $n = 3$ biologically independent replicates. “mock”: non-infected controls. (C) RT-qPCR quantification of ReoT3D RNA in infected U87MG and 293FT cells at MOI 1 at 72 hpi (mean $\pm$ SEM, $n = 3$) (D) U87MG or 293FT cells were infected with MOI of 1 ReoT3D virus. At 72 hpi, cells were lysed and viral titers determined via plaque assay [mean $\pm$ SEM, $n = 3$, $***P < 0.001$, $****P < 0.0001$; significance determined via one-way ANOVA with post-hoc Dunnett’s multiple comparisons test for (C) and (D)]. (E) Time course of VSV-EBOV-GP infection of 293FT cells wild-type and LYSET KO (mean $\pm$ SEM, $n = 3$). (F) Bar-graph depicting independent infections of A549-ACE2 cells $\pm$ LYSET KO with SARS-CoV-2-mNeon using flow cytometry (mean $\pm$ SD, $n = 3$, $****P < 0.0001$; significance determined via unpaired, two-tailed student’s t-test). (G) Infection of SARS-CoV-2 Delta and Omicron Spike reporter virus particles in Huh7.5.1 cell lines with or without LYSET KO. Cells were engineered to stably express ACE2 or ACE2 in combination with TMPRSS2, as indicated. Luciferase activity was measured at 72 hpi and normalized to wild-type cells (mean $\pm$ SEM, $n = 6$, $***P < 0.001$, $****P < 0.0001$; significance determined via significance was determined by unpaired, two-tailed student’s t-test with Welch’s correction).

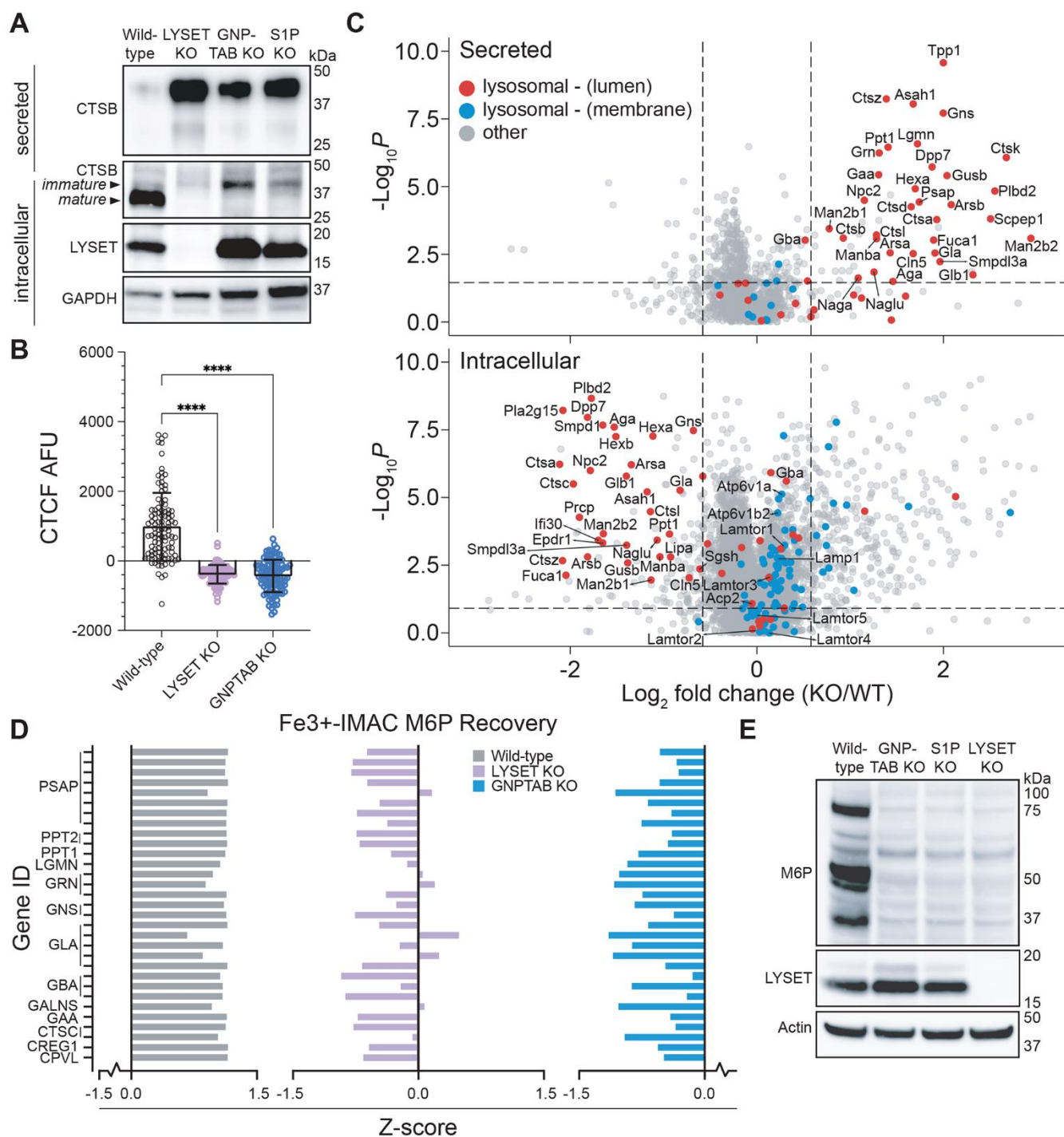


Fig. 2. LYSET is required for global lysosomal enzyme transport via M6P-tagging. (A) Immunoblot analysis of cathepsin B (CTSB) in lysates (intracellular) and in extracellular medium (secreted) from wild-type cells and clonal 293FT cell lines containing knockout (KO) mutations in indicated genes. (B) Cathepsin activity in cells was determined by quantification of BMV-109 fluorescence signal in live 293FT wild-type and KO cells as the raw corrected total cellular fluorescence (CTCF) in arbitrary fluorescence units (AFU) (mean $\pm$ SD, $n = 100$ cells, **** $P < 0.0001$; significance determined via one-way ANOVA with a post-hoc Dunnett's multiple comparisons test). (C) Proteomic analysis of wild-type and LYSET KO MEFs by unbiased data independent acquisition (DIA). DIA was used for intracellular and secreted proteins. (D) Z-score analysis of individual peptides that contain the M6P moiety as determined using glycoproteomics. Peptides were derived from indicated lysosomal proteins in wild-type, LYSET KO and GNPTAB KO 293FT cells; $n = 3$ replicates for each cell line. (E) Immunoblot analysis of M6P-tagged proteins from 293FT wild-type, GNPTAB KO, S1P KO, and LYSET KO cells using an M6P-specific single-chain antibody fragment (M6P).

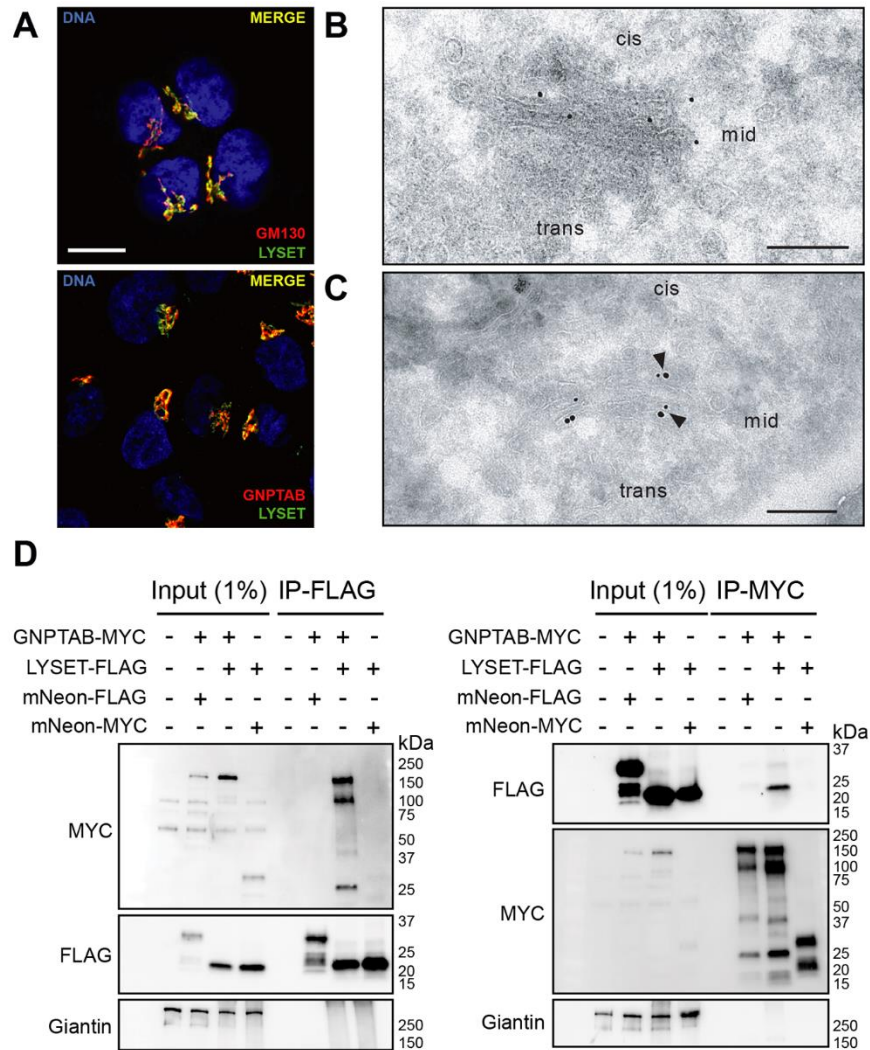


Fig. 3. LYSET binds to GNPTAB and co-localizes with GNPTAB/GNPTG in Golgi apparatus cisternae. (A) Immunofluorescence microscopy in HAP1 cells showing co-localization of LYSET with the cis-Golgi marker GM130, and GNPTAB. Scale bar, 10 μ m (B) Transmission electron microscopy (TEM) immunogold staining (15 nm gold) shows LYSET in Golgi cisternae of SK-MEL-30 wild-type cells using ultrathin sections. (C) TEM double immunogold staining on ultrathin sections for LYSET (15 nm gold) and GNPTG (10 nm gold) in the Golgi apparatus of SK-MEL-30 wild-type cells. Arrowheads indicate co-localization. Scale bar, 200 nm. (D) Immunoprecipitation (IP) on lysates of cells expressing epitope-tagged proteins using anti-FLAG (left panel) or anti-MYC (right panel) magnetic beads, followed by immunoblot analysis with indicated antibodies. Input lysates are also analyzed.

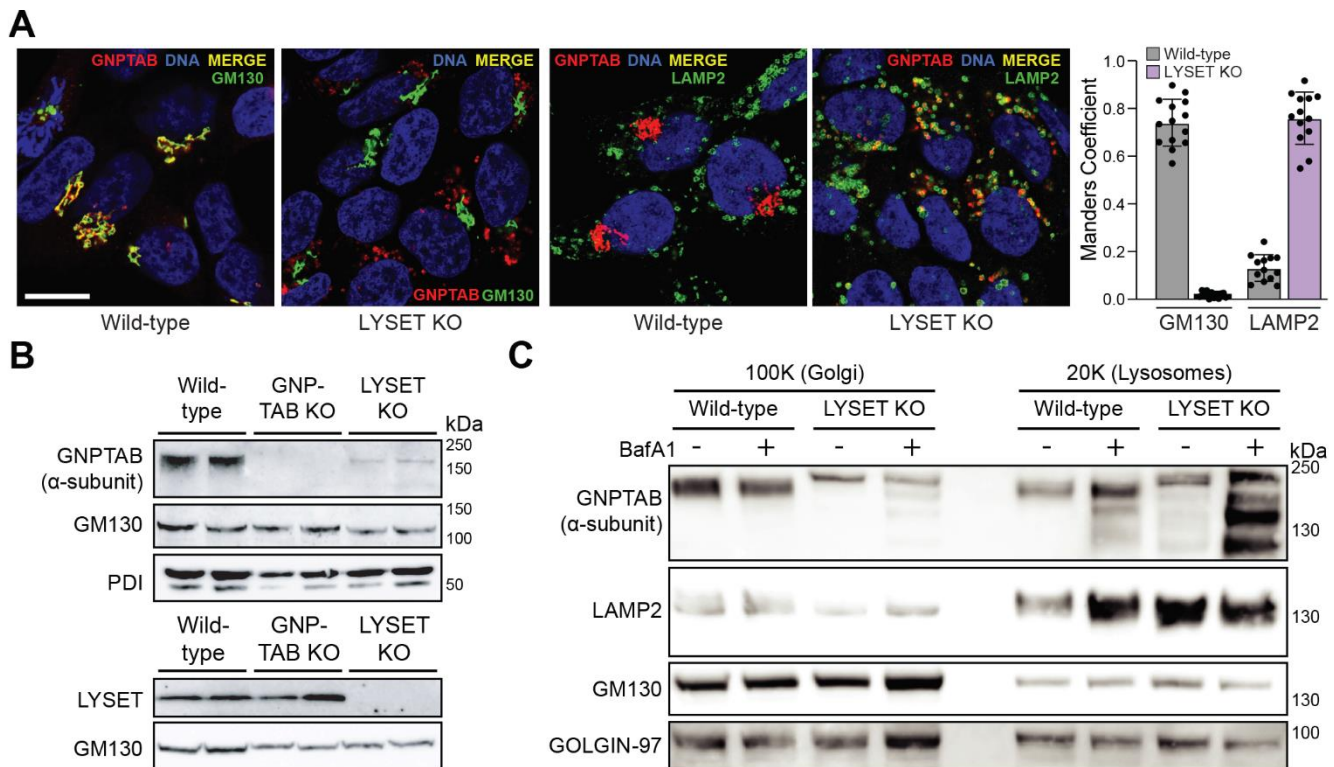


Fig. 4. LYSET KO triggers mislocalization of GNPTAB to the lysosome where it is degraded. (A) Colocalization of GNPTAB with GM130 and LAMP2 in wild-type and LYSET KO cells and Manders colocalization quantification. Colocalization analysis was performed on at least $n = 12$ separate micrograph images. Scale bar, 10 μm **(B)** Immunoblot analysis of GNPTAB in 100K enriched ER/Golgi-fractions from wild-type, GNPTAB KO, and LYSET KO HAP1 cells. The antibody is specific for the α -subunit domain of GNPTAB. Golgi-marker (GM130) and ER-marker (PDI) proteins were used as loading controls. **(C)** Immunoblot analysis of 20K and 100K fractions from wild-type and LYSET KO HAP1 cells with or without bafilomycin A1 (BafA1) treatment. LAMP2, GM130 and GOLGIN-97 proteins were used as controls for preparation and loading.

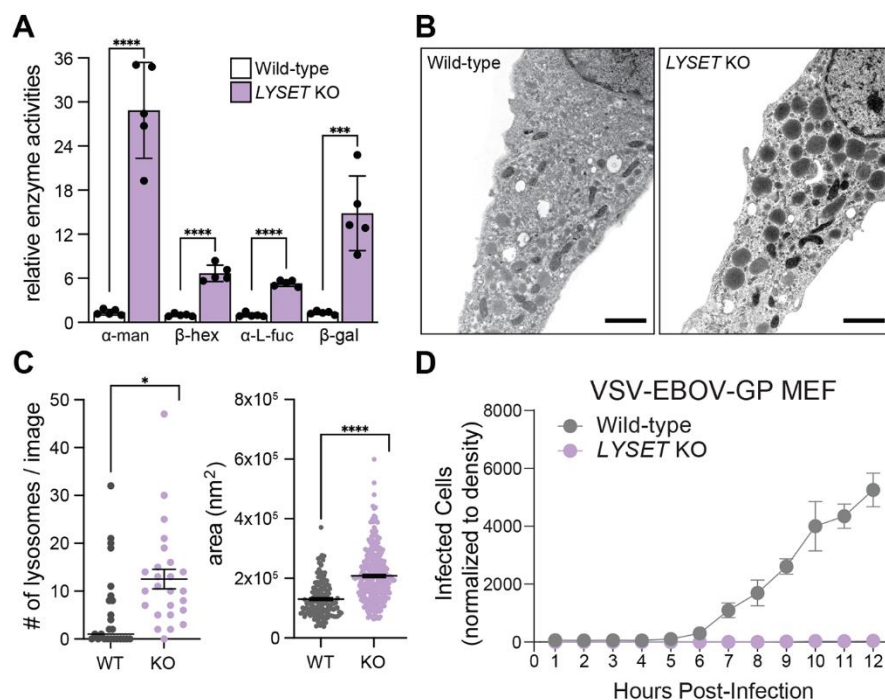


Fig. 5. Lyset KO mice display characteristics used to diagnose ML-II disease including elevated blood serum levels of lysosomal enzymes and aberrant lysosomes in isolated cells. (A) Relative enzyme activities of α -mannosidase (α -man), β -hexosaminidase (β -hex), α -L-fucosidase (α -L-fuc) and β -galactosidase (β -gal) from blood sera of adult wild-type (WT) and *Lyset* KO mice; WT activities set to 1; mean $\pm$ SD, $n = 5$ mice, *** $P < 0.001$, **** $P < 0.0001$, unpaired student's t-test (two-tailed). **(B)** Electron micrographs of WT and *Lyset* KO MEFs. Scale bar, 1 μ m. **(C)** Numbers of lysosomes and areas counted for 25 independent images for WT and *Lyset* KO (KO) MEFs; mean $\pm$ SEM; unpaired, two-tailed student's t-test with Welch's correction; * $P < 0.05$, **** $P < 0.0001$. **(D)** Time course of VSV-EBOV-GP infection of WT and KO MEFs (mean $\pm$ SD, $n = 3$)

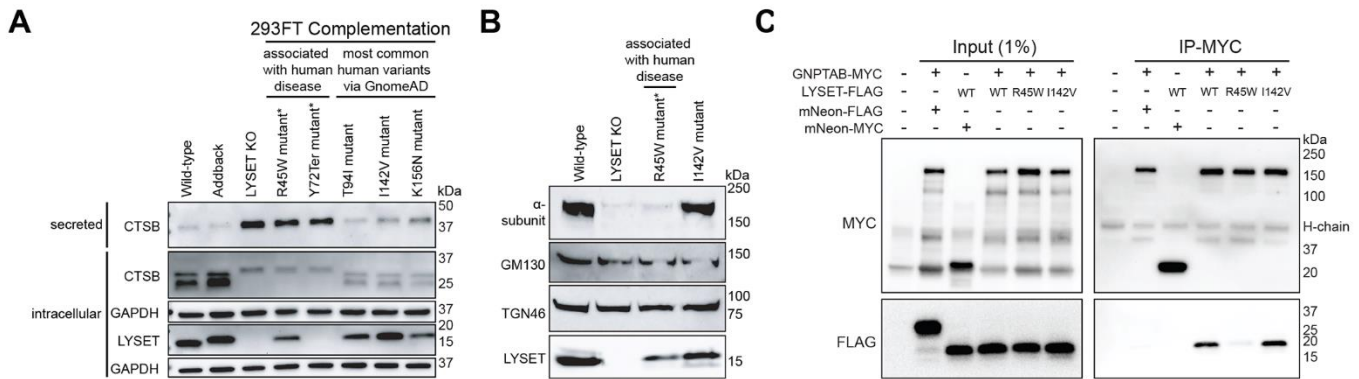


Fig. 6. Pathogenic LYSET mutations fail to restore lysosomal transport defects in LYSET KO cells. (A) Immunoblot of 293FT lysates or supernatants from LYSET KO cells complemented with wild-type LYSET (Addback) or human variants. R45W and Y72Ter are patient-mutations (22); T94I, I142V, and K156N variants are not disease-associated. (B) Immunoblots of 100K enriched-Golgi/ER fractions from 293FT KO cells complemented with indicated LYSET alleles. (C) Immunoprecipitation (IP) on lysates of cells expressing epitope-tagged proteins using anti-MYC magnetic beads followed by immunoblot analysis with indicated antibodies.

The human disease gene LYSET is essential for lysosomal enzyme transport and viral infection

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